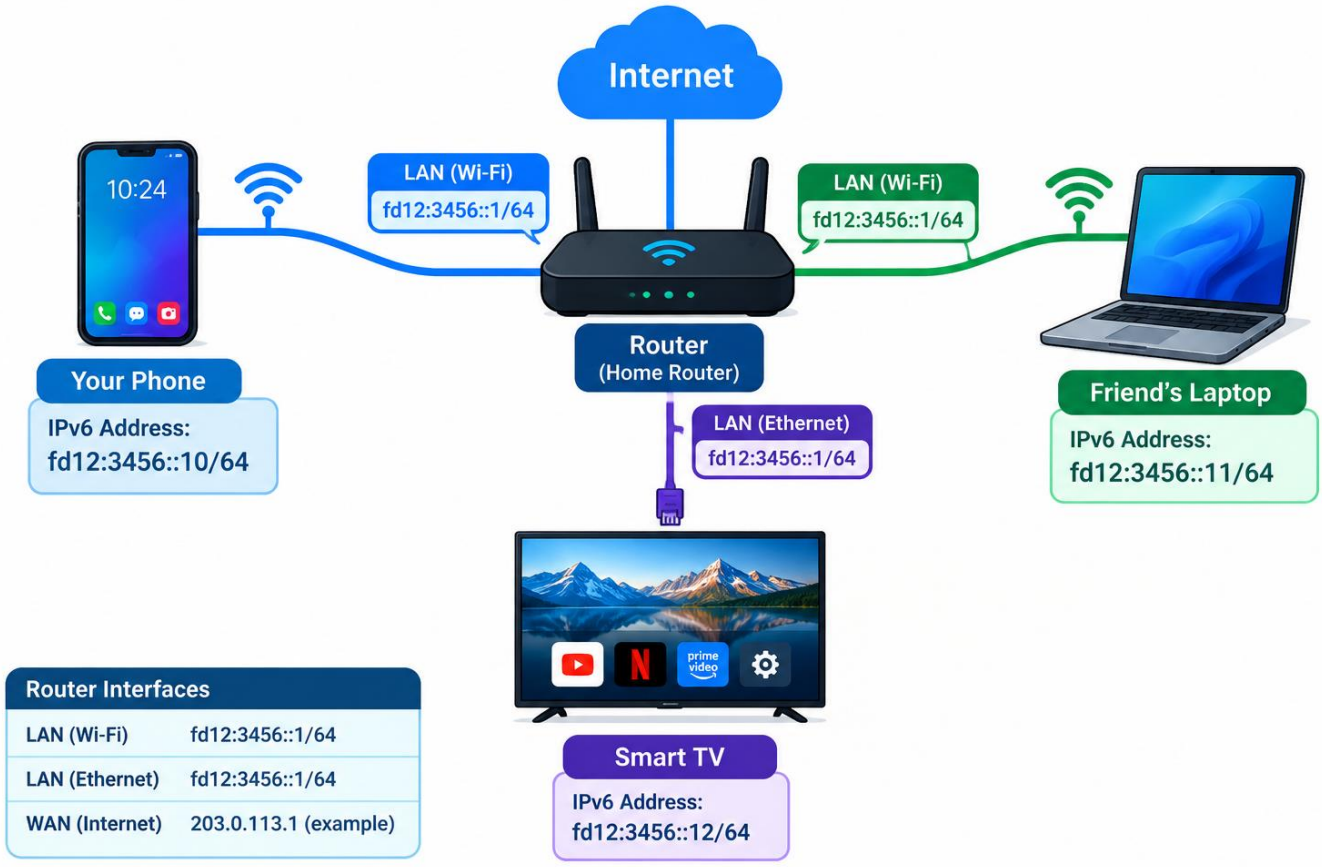
10/13/2026

Assignment-9

Created BY
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1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.Hint:Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Example: How a Packet Travels Between Two Subnets

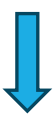
1. Network Setup

- My Laptop IP: 192.168.1.10
- Friend's Computer IP: 192.168.2.20

Both computers are in different subnets, so the packet needs a router.

My Laptop

192.168.1.10



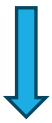
Router R1

192.168.1.1



Router R2

192.168.2.1



Friend's Computer

192.168.2.20

2. Step-by-Step Packet Journey

1. My laptop wants to send data to 192.168.2.20.
2. It checks the destination IP and finds that it is in a different subnet.
3. The laptop sends the packet to its default gateway, Router R1 (192.168.1.1).
4. R1 checks its routing table and forwards the packet toward Router R2.
5. R2 checks its routing table and finds the destination network 192.168.2.0/24.
6. R2 forwards the packet to the friend's computer (192.168.2.20).

3. Routing Table Entries

Router R1

Destination Network	Next Hop
192.168.1.0/24	Directly connected
192.168.2.0/24	Router R2

Router R2

Destination Network	Next Hop
192.168.2.0/24	Directly connected
192.168.1.0/24	Router R1

4. How Does the Router Decide?

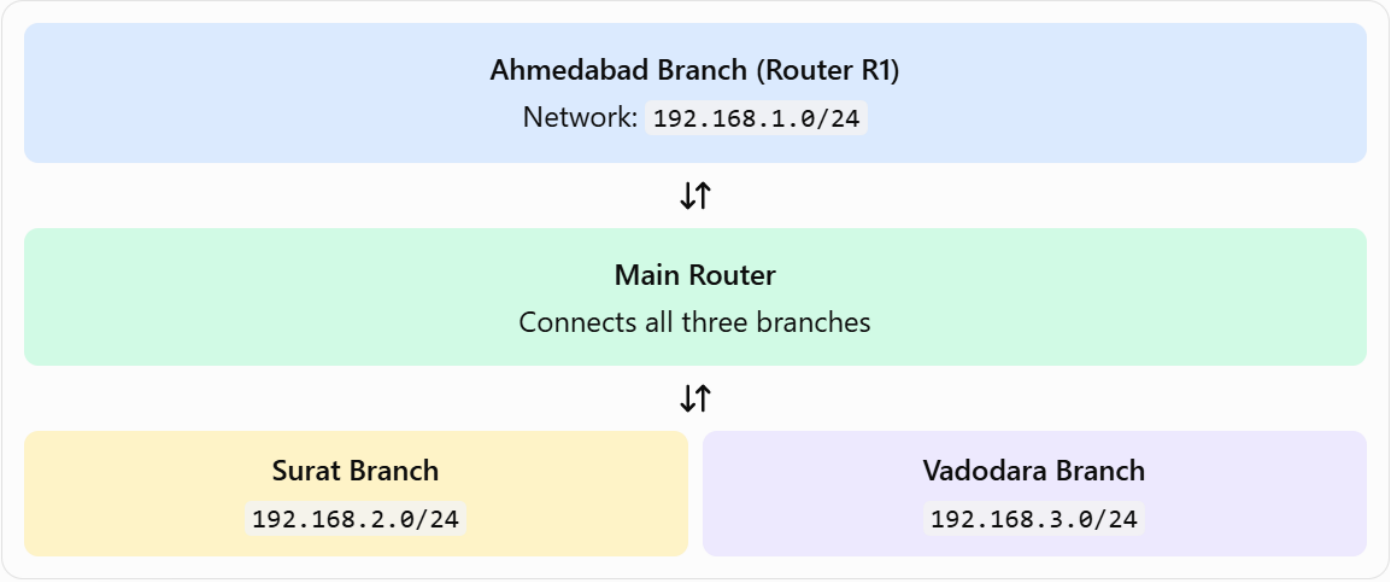
The router checks the destination IP address against its routing table and selects the most specific matching route. It then forwards the packet to the next hop or directly to the destination

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Routing Table for a Zomato-Style Delivery Network

Suppose a food delivery company has three branches in different cities: Ahmedabad, Surat, and Vadodara. Each branch has a different subnet.

1. Network Setup



2. Routing Table

Example routing table for the Main Router:

Network Address	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Directly connected	G0/0
192.168.2.0	255.255.255.0	10.0.0.2	G0/1
192.168.3.0	255.255.255.0	10.0.0.6	G0/2

All IP addresses and interfaces are fictional examples.

3. **How It Works**

   1. **A delivery partner's device sends data to another branch.**
   2. **The router checks the destination network in its routing table.**
   3. **It selects the matching route.**
   4. **It forwards the packet through the specified interface to the next hop.**
   5. **The destination branch router delivers the packet to the correct device.**

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

❖ Static Routing vs Dynamic Routing

Point	Static Routing	Dynamic Routing (RIP / OSPF)
Meaning	Admin manually adds routes.	Routers learn routes automatically using protocols.
3 Pros	1. Easy for small networks. 2. No routing protocol traffic. 3. More control over routes.	1. Automatically learns routes. 2. Can find another route after a failure. 3. Good for large networks.
3Cons	1. Manual configuration takes time. 2. Routes do not automatically adjust to failures. 3. Difficult to manage large networks.	1. More complex to configure. 2. Uses CPU, memory, and network resources. 3. Can take time to learn a new route after a failure.

Example	A small restaurant uses a fixed route to its head office.	A food delivery company uses OSPF to connect branches across different cities.

Real-World Example: Food Delivery App

- Static Routing: A restaurant branch sends data to the main office through a fixed route. If that route fails, an administrator may need to change it.
- Dynamic Routing: If a network link between branches fails, OSPF can learn an alternative route and redirect network traffic.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

OSPF Configuration Between Two Routers (Cisco)

1. Network Setup

- Router R1 LAN: 192.168.1.0/24
- Router R2 LAN: 192.168.2.0/24
- Router-to-router link: 10.0.0.0/30

2. Router R1 Configuration

```
enable
configure terminal
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
```

Explanation:

- enable — Enters privileged mode.
- configure terminal — Enters configuration mode.
- router ospf 1 — Starts OSPF process 1.

- network 192.168.1.0 0.0.0.255 area 0 — Enables OSPF on the R1 LAN interface.
- network 10.0.0.0 0.0.0.3 area 0 — Enables OSPF on the router-to-router link.
- end — Exits configuration mode.

3. Router R2 Configuration

```
enable
configure terminal
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
```

Explanation:

- router ospf 1 — Starts the OSPF process.
- network 192.168.2.0 0.0.0.255 area 0 — Enables OSPF on the R2 LAN interface.
- network 10.0.0.0 0.0.0.3 area 0 — Enables OSPF on the router-to-router link.
- end — Exits configuration mode.

4. Verify the Configuration

Run this command on both routers:

```
show ip ospf neighbor
```

Conclusion

OSPF allows R1 and R2 to exchange network routes automatically. Once the interfaces are correctly configured with IP addresses and are active, PCs in 192.168.1.0/24 can communicate with PCs in 192.168.2.0/24, provided their gateways and other required settings are correct.